

CHAPTER 5

*Conductive Polymers
and Ladder Polymers*

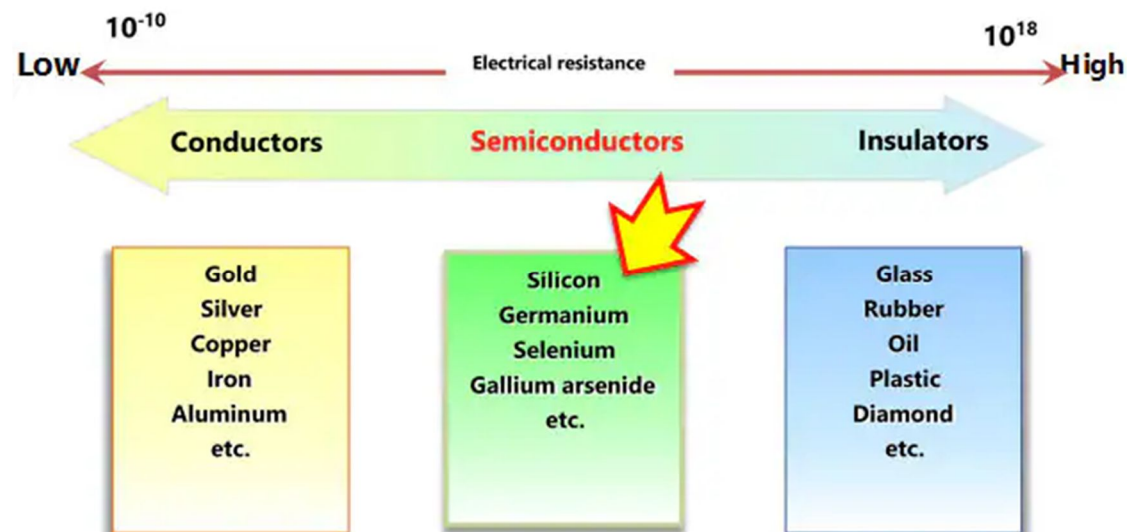
Material	Conductivity σ [$\text{S}\cdot\text{cm}^{-1}$]
Isolator	$< 10^{-7}$
Semiconductor	$10^{-7}-10^2$
Metal	$> 10^2$

$$\sigma = N \cdot e \cdot \mu \quad [\text{S}\cdot\text{m}^{-1}]$$

N = number of charge carriers per unit volume [m^{-3}]

e = charge of the carrier [C]

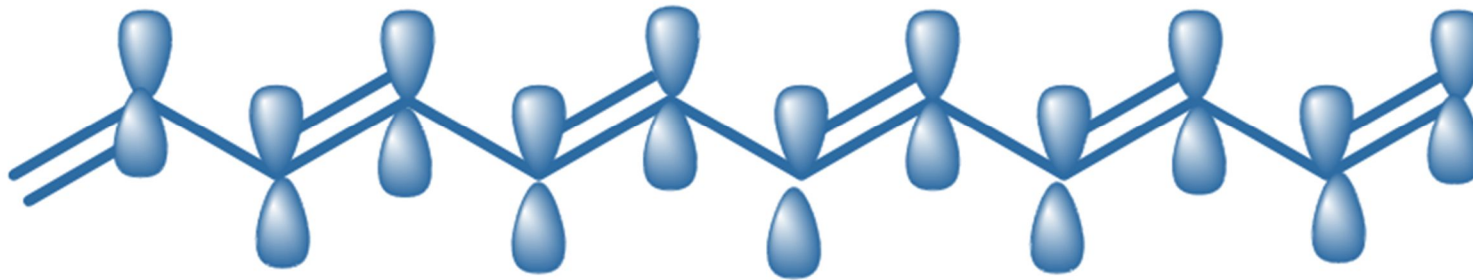
μ = velocity of the charge carrier [$\text{cm}^2\cdot\text{V}^{-1}\cdot\text{s}^{-1}$]



Introduction

- A large part of commodity polymers are isolators, where both N and μ are small (e.g., for nylon 66, PE, PTFE, $\sigma < 10^{-12} \text{ S}\cdot\text{cm}^{-1}$).
- (Semi-)conducting polymers, where N is small but μ is large, also exist.

Requirement: extended conjugated system along the backbone (at least, 30 monomer units)



Simplest structure: Polyacetylene

Polyacetylene and Historical Development

Polyacetylene is the first polymer, for which electrical conductivity was observed.

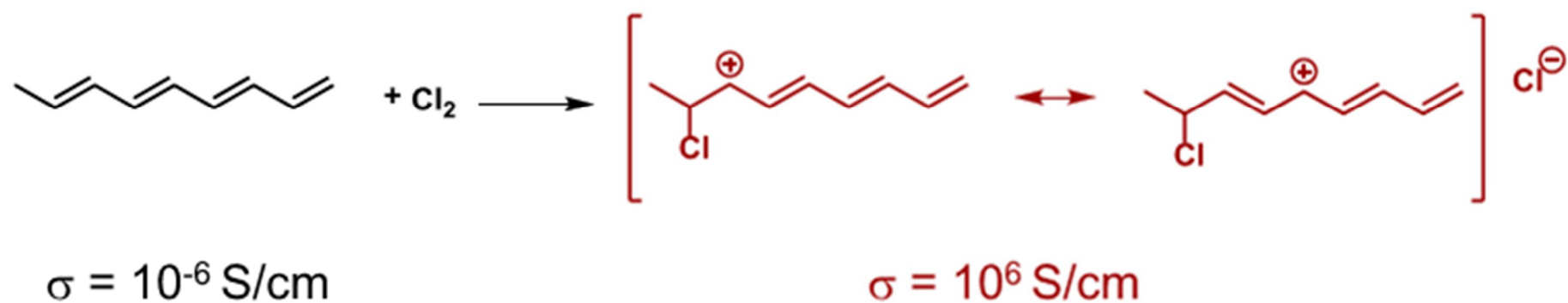
- **1958** Natta (IT) polymerized acetylene for the first time to **polyacetylene**.
- **1963** Weiss (AUS) discovered the conductivity of **polypyrrole** obtained by electropolymerization, particularly in the presence of adsorbed iodide.
- **1976** Heeger (US), MacDiarmid (US), and Shirakawa (JP) demonstrated that **polyacetylene**, when doped with iodine, could exhibit electrical conductivity comparable to metals.
- **1980s** The field rapidly expanded and “new” polymers were explored (polyaniline, polypyrrole, polythiophene).
- **1990s** Conducting polymers found applications in electronic devices, including: organic light-emitting diodes (OLEDs), electrochromic displays, antistatic coatings, as well as sensors and actuators. The field of organic electronics emerged, combining materials science and polymer chemistry.

- **2000** Heeger, MacDiarmid, and Shirakawa were awarded the Nobel Prize in Chemistry for their discovery and development of conducting polymers.

Picture of a *trans*-polyacetylene sample provided by Professor Alan Heeger, October 2001 (picture courtesy of Demetrio Filho)



- **2000s–now** Conducting polymers have been integrated into flexible electronics, bioelectronics, and energy storage devices (e.g., supercapacitors, batteries).
Introduction of nanotubes and nanowires improved performance in many applications and biocompatibility was placed in focus for use in neural interfaces and tissue engineering.

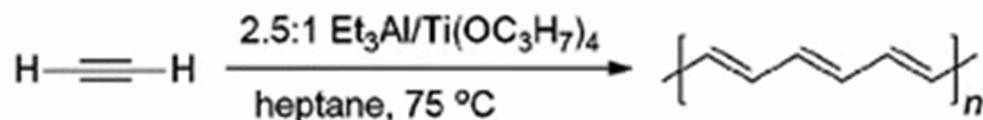


Conductive Polymers

Polyacetylene

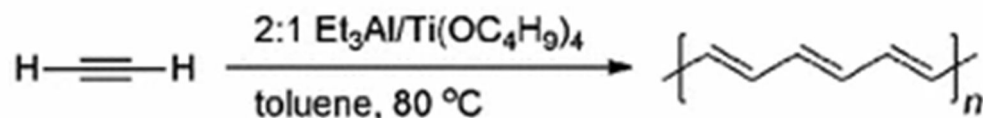
Historical synthetic method: Insertion polymerization with Ziegler-Natta catalysts

Natta's conditions:



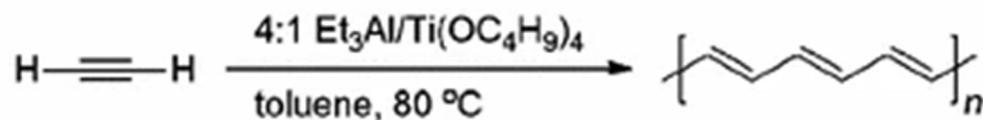
black insoluble material

Hanato's conditions:



black insoluble material

Pyun and Shirakawa's conditions:



lustrous, silvery film; „ragged pieces of a film“

approx. 1000 times more catalyst (by accident!)

Rasmussen, S.C. (2018). Polyacetylene. In: *Acetylene and Its Polymers. SpringerBriefs in Molecular Science. Springer, Cham.*

Polyacetylene

Shirakawa's method: Walls of reaction flask coated with the Ziegler–Natta catalyst and acetylene introduced as gas → immediate formation of a film

Reaction parameters (particularly temperature) influence the exact structure.

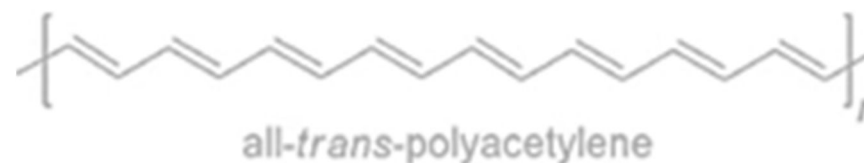
Polymer films prepared at temperatures below $-78\text{ }^{\circ}\text{C}$ were described to have a “copper-like luster”.



*Thermal isomerization at $200\text{ }^{\circ}\text{C}$ for 10 min
(starts at $75\text{ }^{\circ}\text{C}$ but slow)*

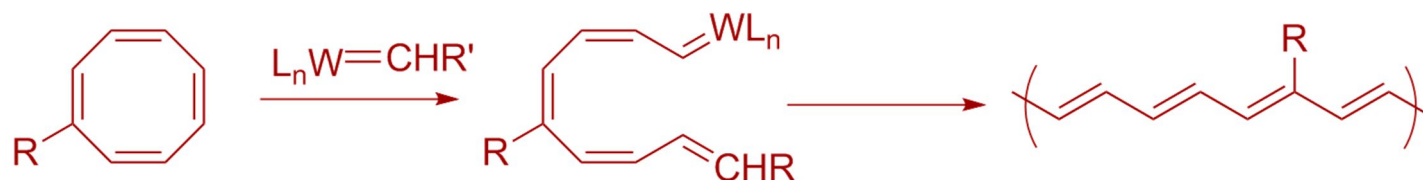


Films prepared above $150\text{ }^{\circ}\text{C}$ were described as being an “intense black material with a metallic luster” or having a “silvery” appearance on the surface.



Rasmussen, S.C. (2018). Polyacetylene. In: *Acetylene and Its Polymers. SpringerBriefs in Molecular Science.* Springer, Cham.

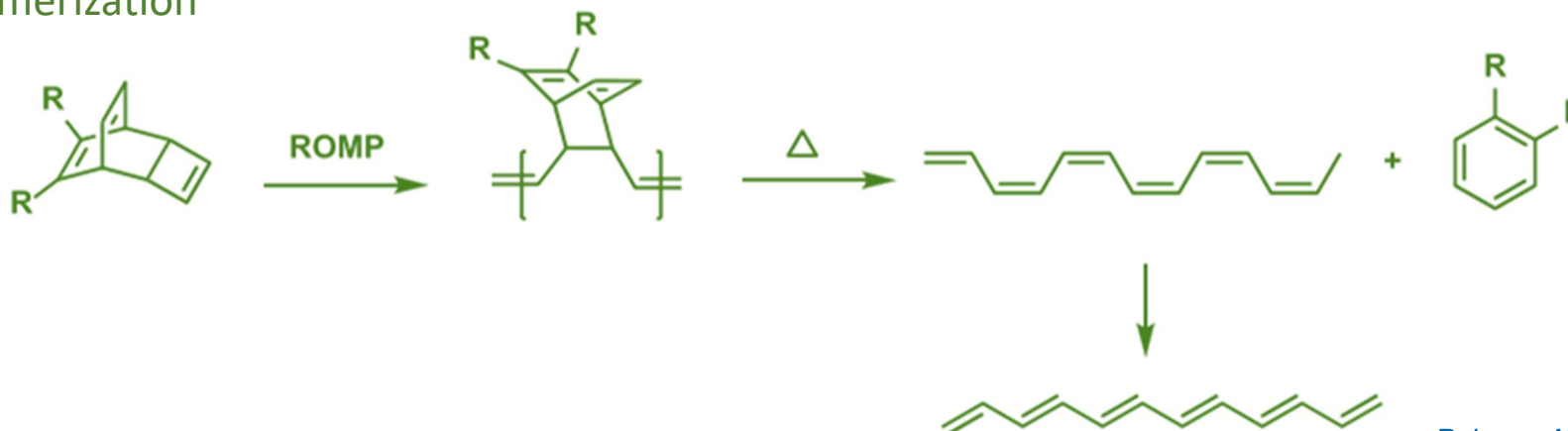
- Radiation (glow discharge, ultraviolet, and γ -radiation)
- ROMP from cyclooctatetraenes



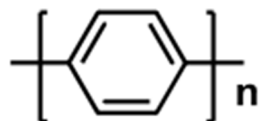
R = linear or branched alkyl group

J. Am. Chem. Soc. **1993**, 115, 1397

- ROMP from 7,8-bis(trifluoromethyl)tricyclo-[4.2.2.0²⁵]deca-3,7,9-triene, followed by cycloreversion and isomerization



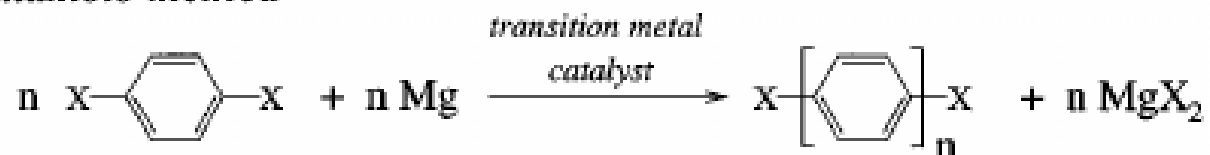
Polymer **1980**, 21, 595



Kovacic method

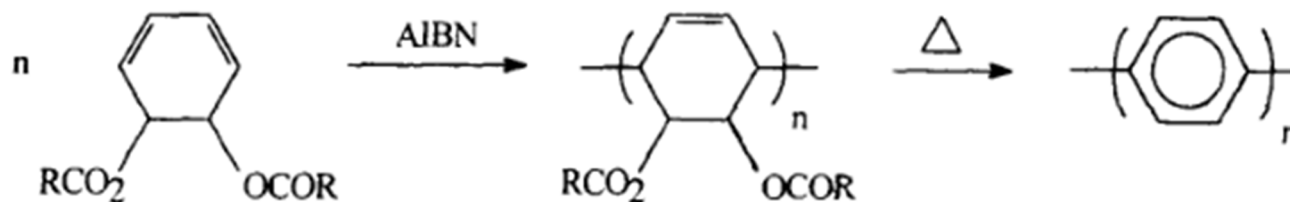


Yamamoto method



X = halogen

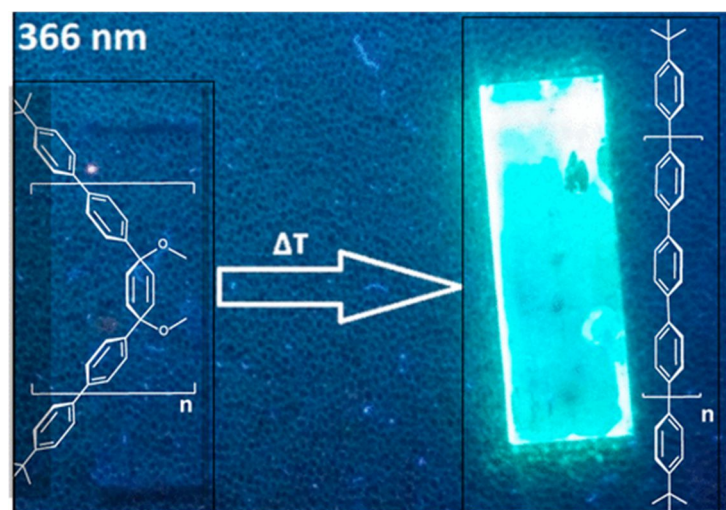
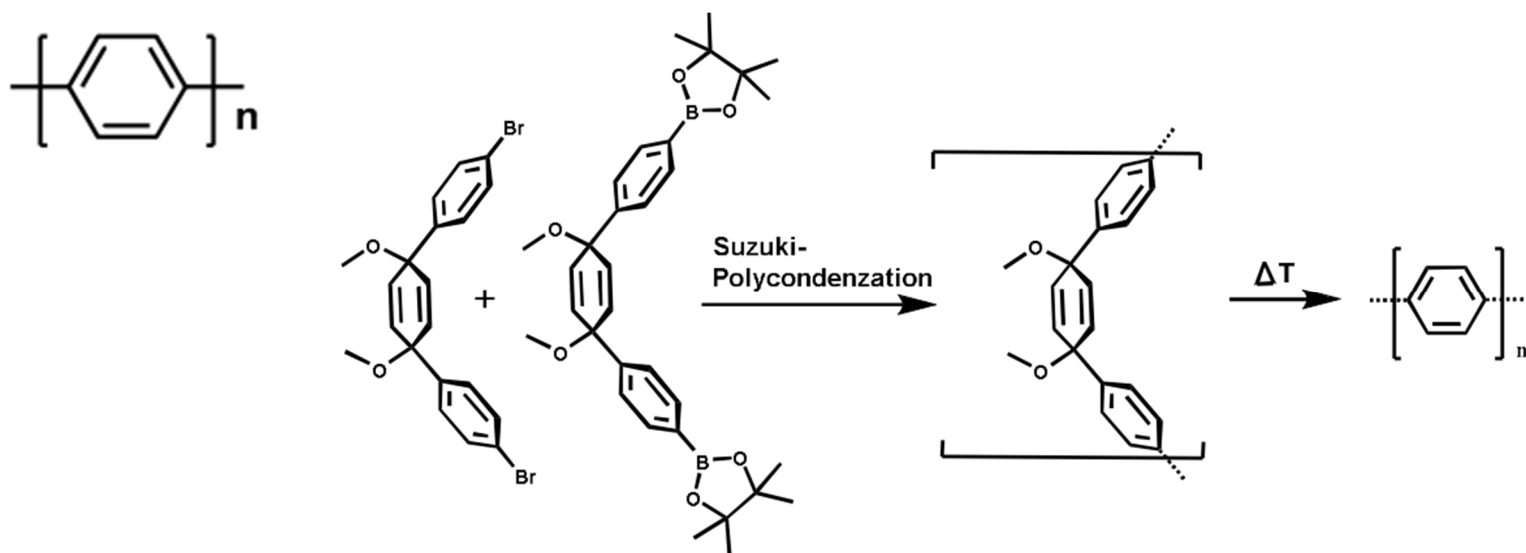
J. Mater. Res. **1999**, 14, 1124



Macromolecules **1987**, 20, 1787

Conductive Polymers

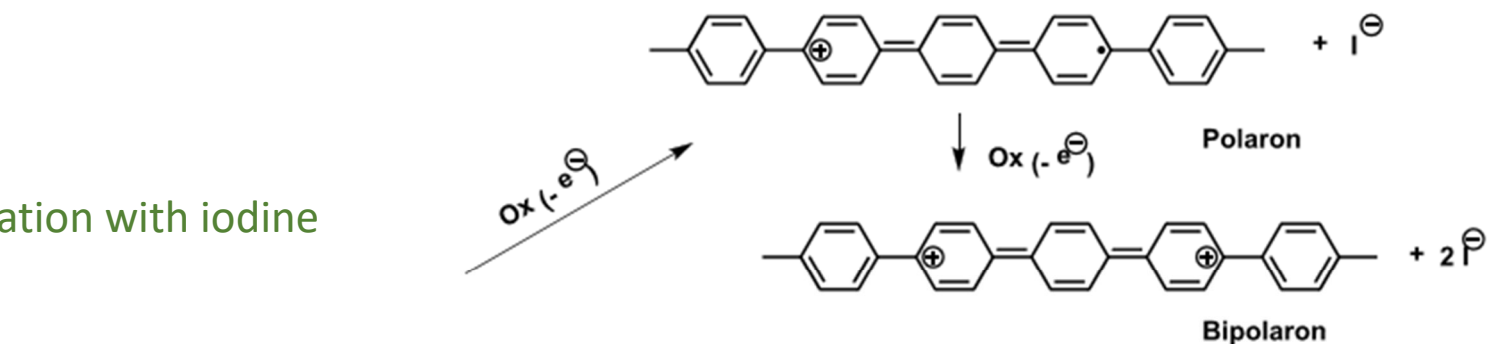
Poly(*p*-phenylene)



Conductive Polymers

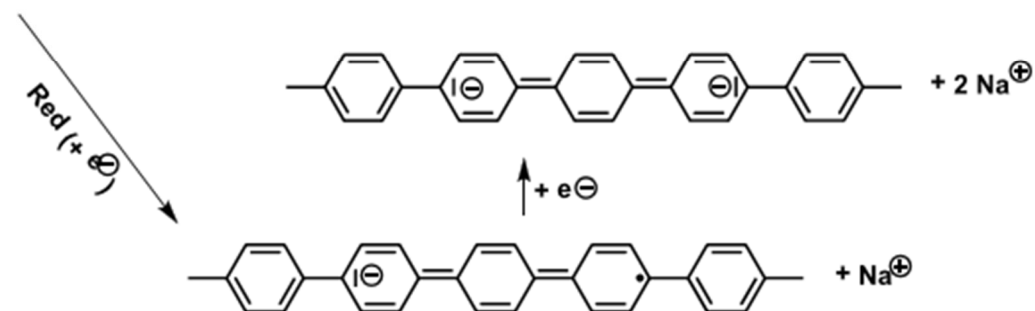
Poly(*p*-phenylene)

Oxidation with iodine



p-type

Reduction with sodium
(potassium)

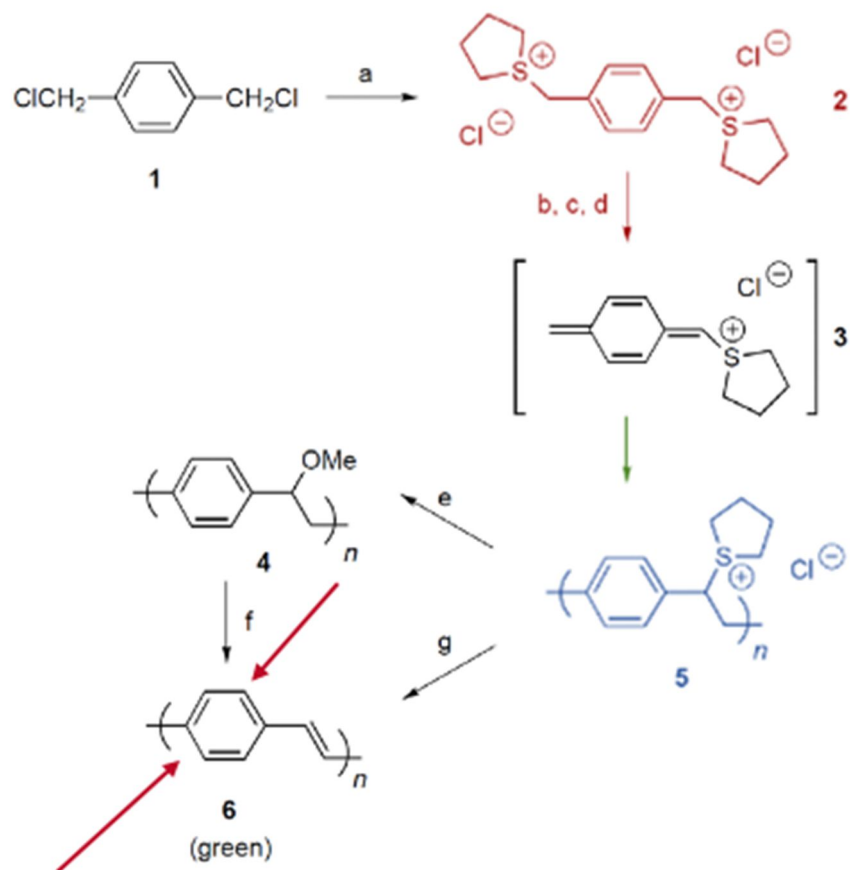


n-type

Conductive Polymers

Poly(p-phenylene vinylene)

Scheme 1. Synthesis of PPV (**6**): a) tetrahydrothiophene, MeOH, 65 °C; b) NaOH, MeOH/H₂O or Bu₄NOH, MeOH, 0 °C; c) neutralization (HCl); d) dialysis (water); e) MeOH, 50 °C; f) 220 °C, HCl(g)/Ar, 22 h; g) 180–300 °C, vacuum, 12 h.

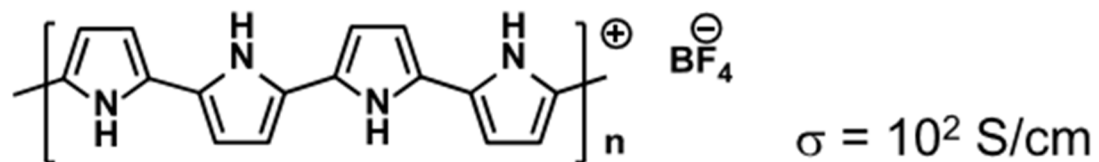


Monomer precursor

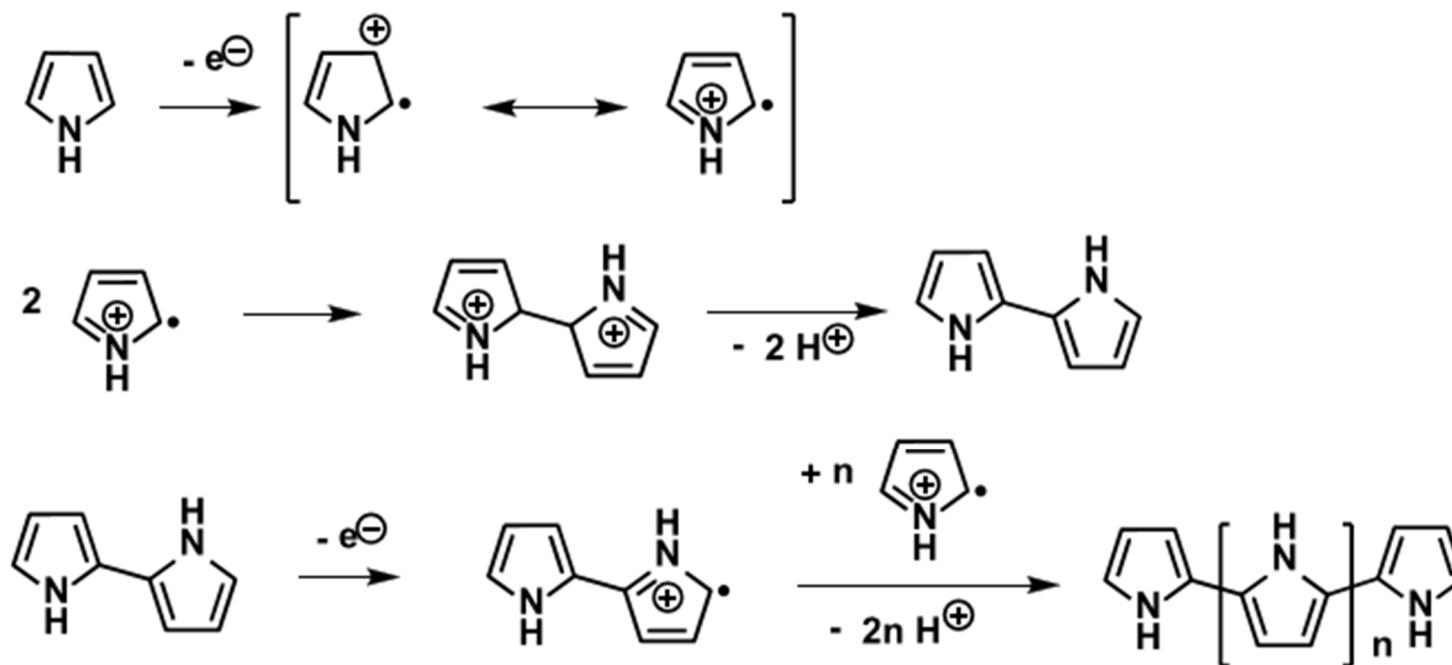
Polymerization (probably radical)

Can be used for coating (spin-coating, drop-casting), then converted by heating

Note: alkoxy-substituted PPVs are organosoluble and can be produced and processed without heating.



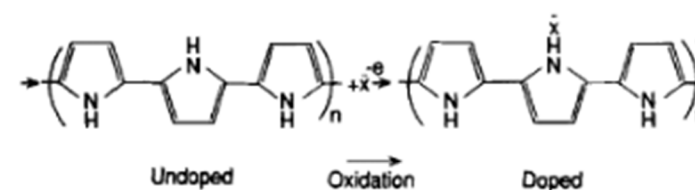
Synthesis via anodic oxidation in the presence of $\text{Et}_4\text{N}^+\text{BF}_4^-$

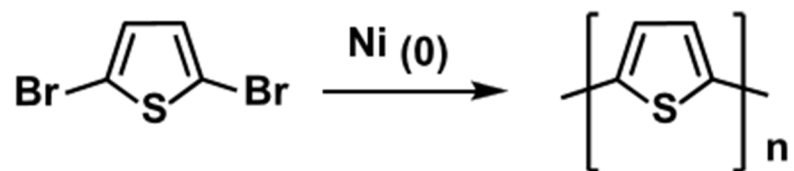


“[...] electrical conductivity, thermal stability, processibility, and electrochemical growth not only depends on oxidation state of a polymer but are also dependent on the nature of counter anion which can be defined in terms of its electronegativity, molecular size, orientation and solvation (Trivedi 1997; Zotti et al 1988).”

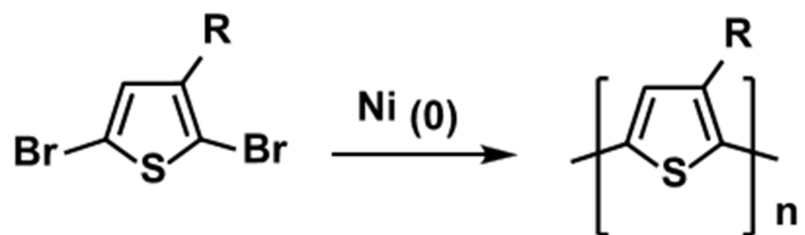
Table 3. Electrochemical data on polypyrrole.

Electrolyte	E_{pa} (mV)	E_{pc} (mV)	Conductivity (S/cm)
<i>In Acetonitrile</i>			
$LiClO_4$	-110	-440	100
Me_4NPF_6	-160	-325	50
Et_4NBF_4	-100	-300	20
Et_4NClO_4	-090	-290	100
Bu_4NBF_4	-110	-260	20
<i>In aq. medium</i>			
Dodecylbenzene sulphonate	+170	-070	01



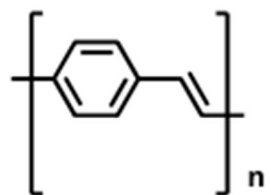


hardly soluble

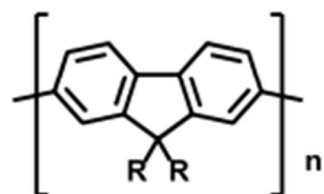


HT-PAT

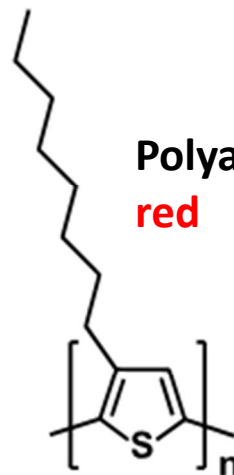




Poly(*p*-phenylene vinylene) (PPV)
yellow-green

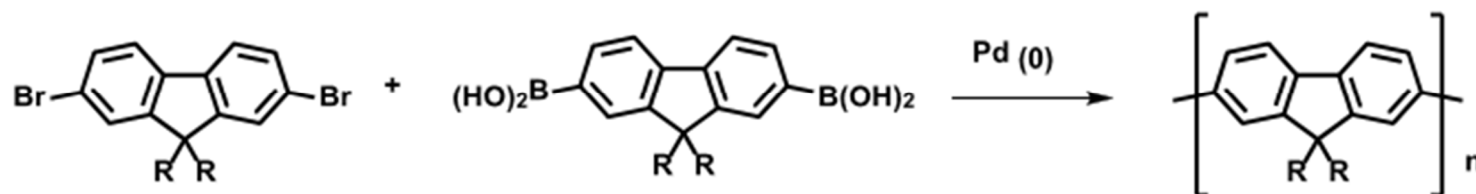


Polyfluorene (PPV)
blue



Polyalkylthiophene (PAT)
red

Synthesis by Suzuki polymerization, AA + BB



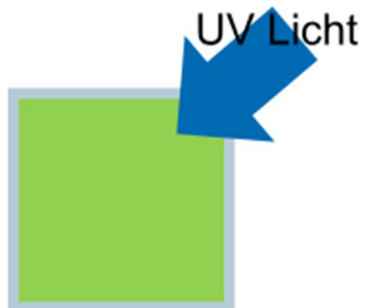
J. H. Burroughes, Richard Friend, et al. in *Nature* 1990

Konjugierte Polymere: Photolumineszenz

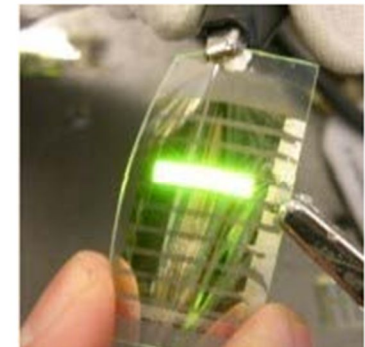
konjugiertes
Polymer

Konjugierte Polymere als Halbleiter/Leiter mit Photolumineszenz: Elektrolumineszenz

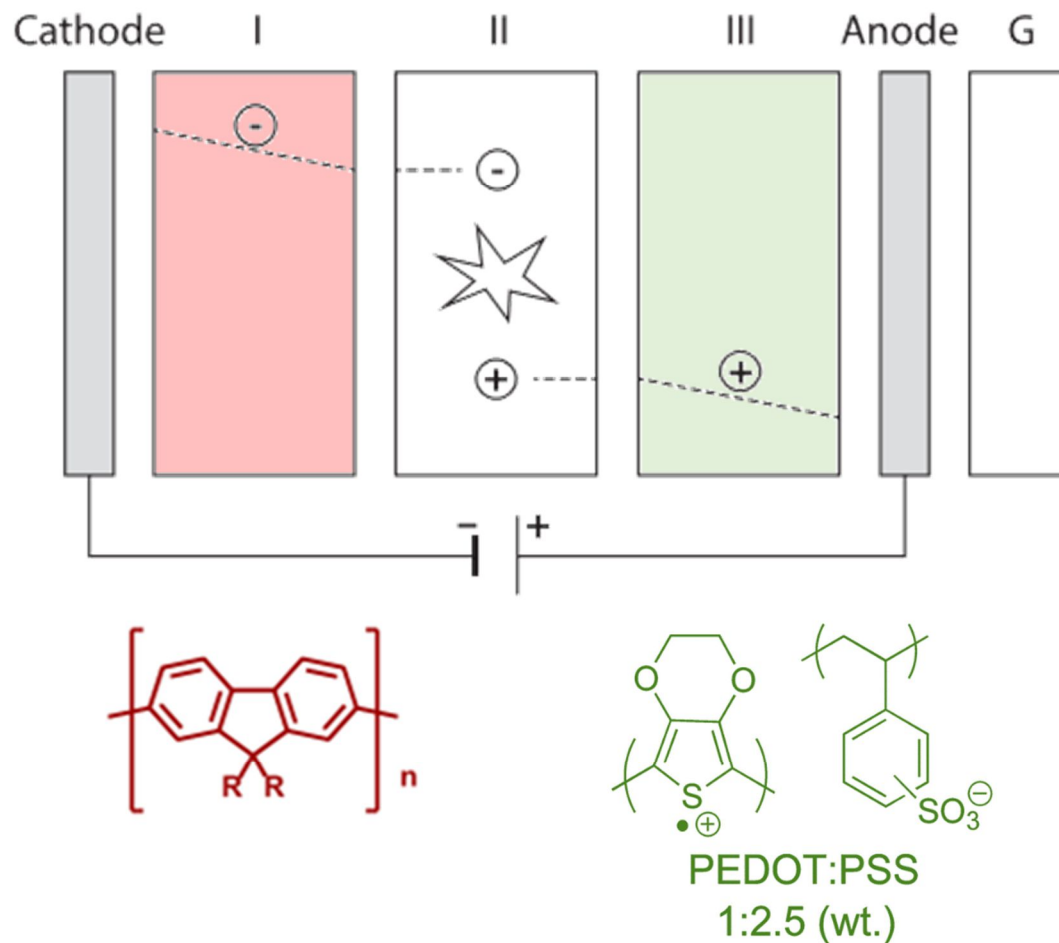
Einbau in eine LED (als Schicht zwischen zwei Elektroden): Polymer LED (PLED), OLED

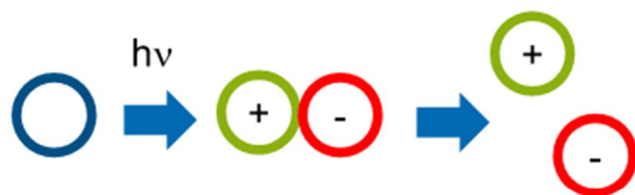
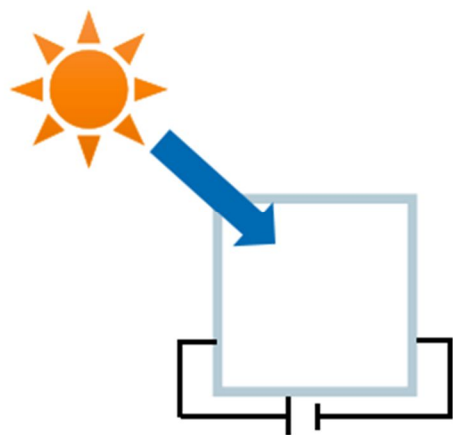


A. Teichler et al./*European Polymer Journal* 49 (2013) 2186–2195



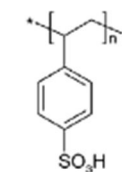
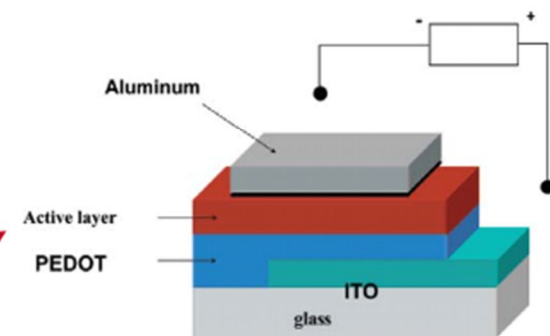
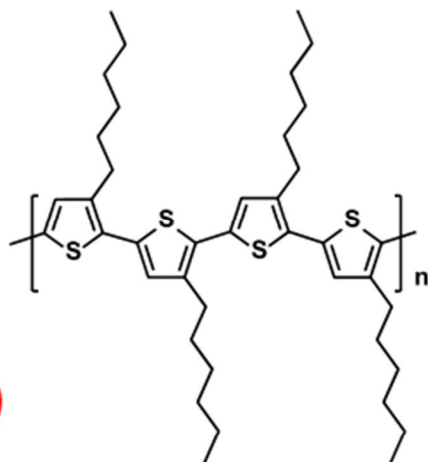
■ Fig. 22.10 Schematic construction of a three-layer OLED. *I* Electron Conductor, *II* Emitter, *III* Hole conductor, *G* Glass support, *Cathode* Ba, Ca, Al, *Anode* ITO (indium tin oxide)



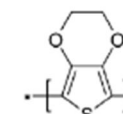


1. Excited state formation (electron-hole pair)
2. Charge separation
3. Diffusion to the electrodes

Poly(3-hexylthiophene) (P3HT)



PSS



pEDOT

p-Typ

Chemical Reviews, 2007, Vol. 107, No. 4



Charge transport along the chain (ordered)



Charge transport between chains (disordered) (energy barrier)

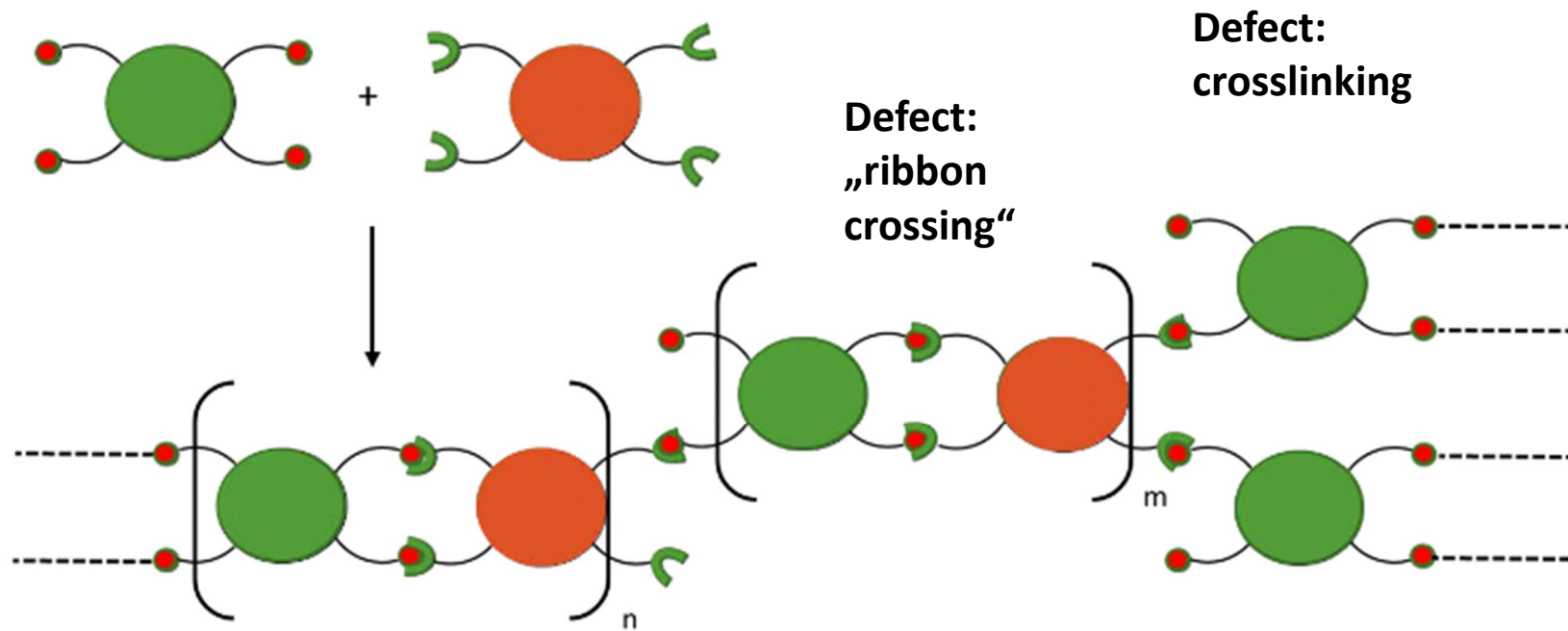
Ordered polymer chains, as long as possible, are the goal for optimal conductivity

→ **Rigidification of the chain with conservation of the conjugation**

Ladder Polymers

Two methods for synthesis

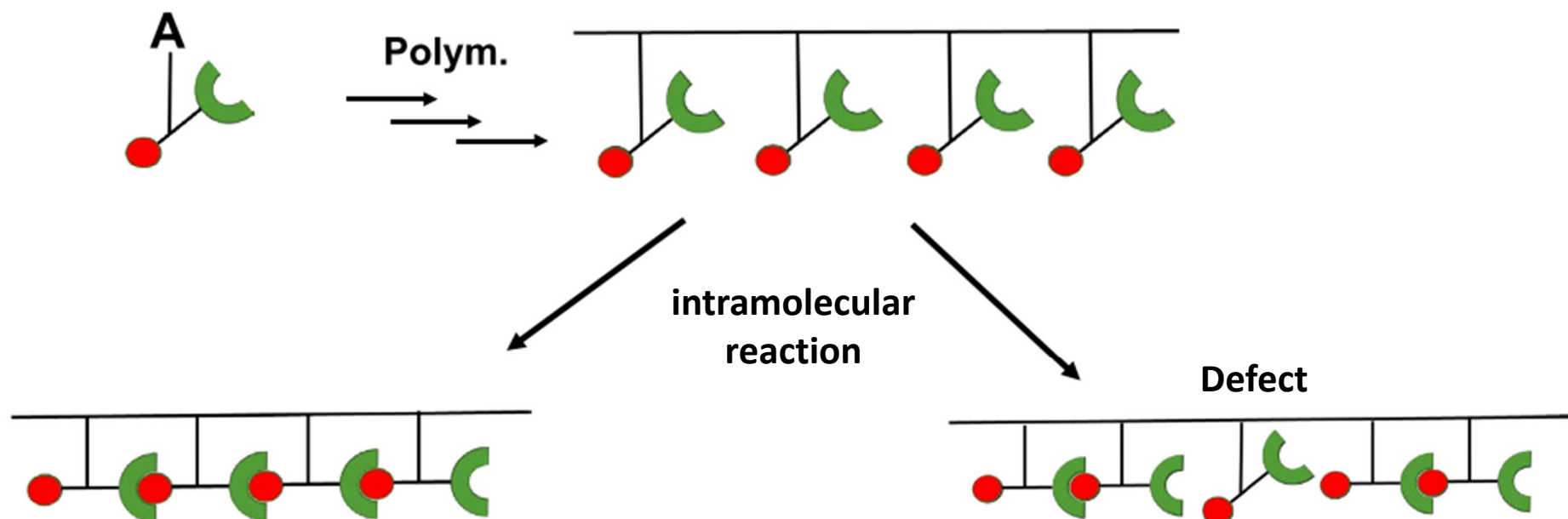
A] Polycondensation/polyaddition



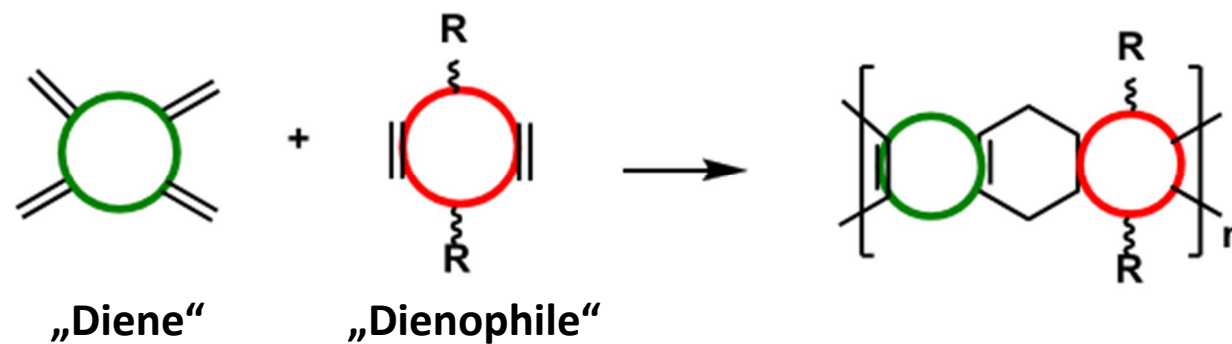
Ladder Polymers

Two methods for synthesis

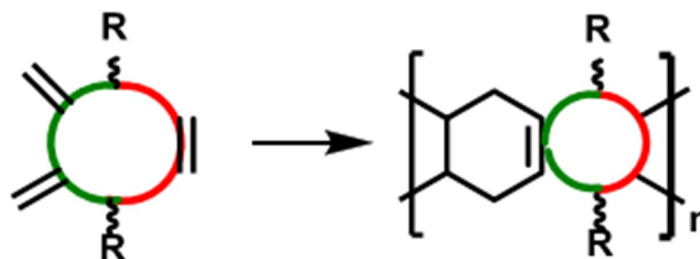
B] Polymer-analogue reaction



1] AA + BB monomers



2] AB monomer

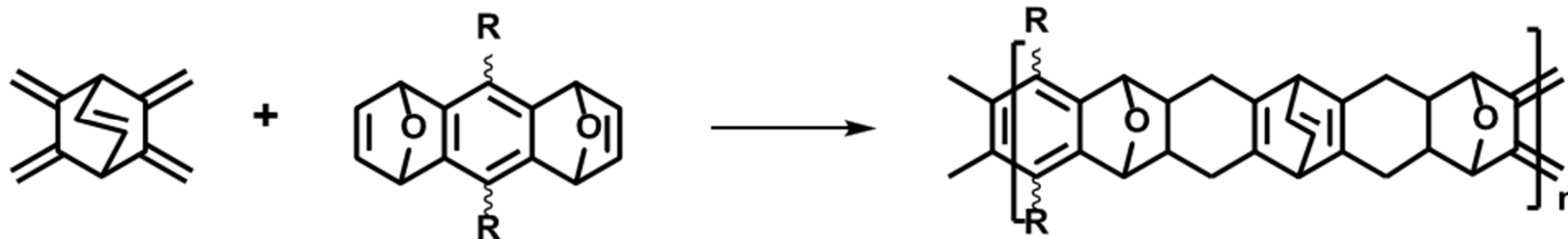


R = alkyl or alkoxy
(for solubility)

Ladder Polymers

Example of AA + BB system

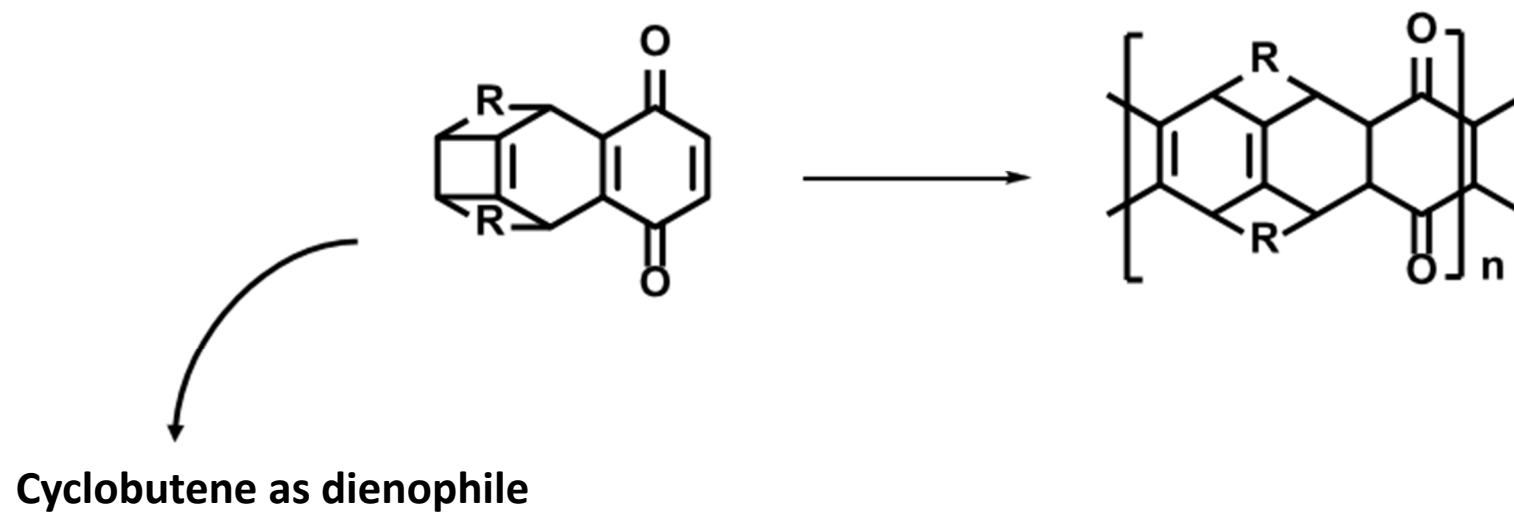
Wegener and Müllen (1991)

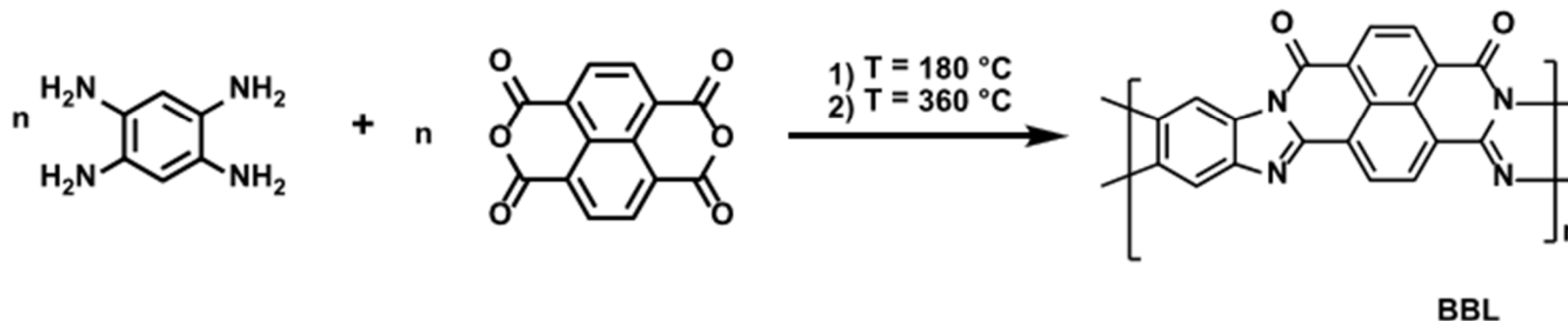


Ladder Polymers

Example of AB system

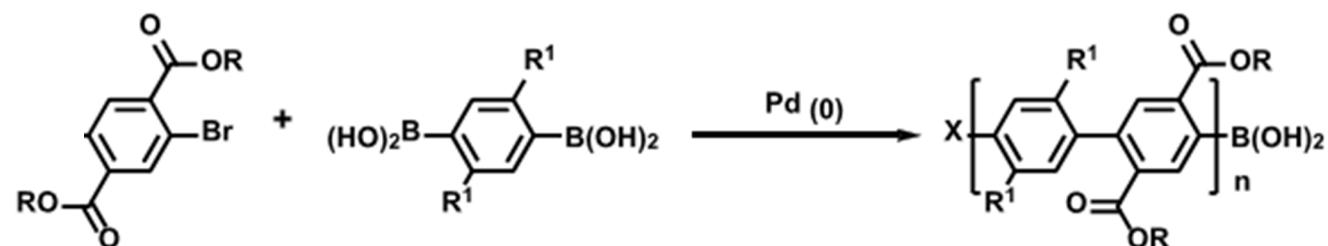
Schlüter (1991)



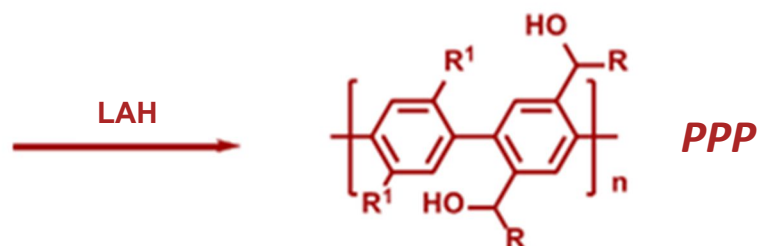


Commercially available polymeric semi-conductor with n-type behavior in organic field-effect transistor (FETs) and photovoltaic cells (PVs). Processed from solutions in methanesulfonic acid.

Suzuki polymerization



Reduction



Friedel–Crafts alkylation

